Magnetic overprinting in the Thanet gabbro complex, Ontario

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Multi-component paleomagnetic remanences are reported for the Thanet gabbro complex of the Grenville Province in Ontario. Forty-five samples (10 sites) attain stable reversed (A) directions in AF's (alternating fields) of 150 Oe (1.2×10^4 A/m) or less. Blocking temperatures (T_B's) are 300–350°C (5 sites) and 525–560° (5 sites). The A_1 paleopole of 158.4°E, 38.0°S ($k = 593$, $\alpha_{95} = 3.1°$) for the low T_B sites differs significantly at the 95% confidence level from the A_2 pole of 172.0°E, 32.3°S ($k = 151$, $\alpha_{95} = 6.2°$) for the high T_B sites. Directions from 18 additional samples (at 8 sites) follow great circle paths towards the A direction upon AF demagnetization to 1000 Oe (8.0×10^4 A/m). T_B's of the lower coercivity (B) component are 560–580°C, *above* T_B's of both A components. Upon thermal demagnetization, directions swing *away* from the A direction and endpoints reached above 560°C define a normal B paleopole at 159.3°E, 20.3°N ($k = 9.6$, $\alpha_{95} = 11.7°$).

If the A and B components are both of thermal origin, B predates A, and was overprinted by A during a regional metamorphic event which reset components with T_B's < 560°C but left those unaffected with T_B's > 560°C. The difference in the A_2 and A_1 directions represents apparent polar wander during cooling from above 500°C to less than 300°C. If the B component, however, was generated as a low temperature (<300°C) chemical remanence, the relative age of A and B are reversed—B is younger than A. The applicaton of Thanet paleopoles to the problems of Grenville and Sveconorwegian tectonics is described.

On rapporte les rémanences paléomagnétiques à plusieurs composantes pour le complex gabbroïque de Thanet dans la province de Grenville de l'Ontario. Quarante-cinq échantillons (10 sites) arrivent à des directions renversées (A) stables pour des champs alternatifs de 150 Oe (1.2×10^4 A/m) ou moins. Les températures de blocage (T_B) sont de 300–350°C (5 sites) et de 525–560°C (5 sites). Le paléopôle A_1 de 158.4°E, 38.0°S ($k = 593$, $\alpha_{95} = 3.1°$) pour les sites à faibles T_B diffère de façon significative au niveau de confiance de 95% du pôle A_2 de 172.0°E, 32.3°S ($k = 151$, $\alpha_{95} = 6.2°$) pour les sites à T_B élevées. Les directions de 18 échantillons additionnels (provenant de 8 sites) suivent des trajets de grand cercle vers la directions de A après désaimantation AF à 1000 Oe (8.0×10^4 A/m). Les T_B de plus faible composante de coercivité (B) vont de 560–580°C, ce qui est *au-dessus* des T_B des deux composantes de A. Après désaimantation thermique, les directions s'éloignent de la direction A et les points limites atteints au-dessus de 560°C définissent un paléopôle normal B à 159.3°E, 20.3°N ($k = 9.6$, $\alpha_{95} = 11.7°$).

Si les composantes A et B étaient toutes deux d'origine thermique, B serait plus ancien que A et aurait subi une surimpression par A au cours de l'épisode de métamorphisme régional qui a réouvert le système des composantes dont T_B <560°C mais a laissé intactes celles dont T_B > 560°C. La différence entre les directions A_1 et A_2 représente la dérive polaire apparente durant le refroidissement de plus de 500°C à moins de 300°C. Si la composante B, toutefois, a été engendrée à basse température (<300°C) comme rémanence chimique, l'âge relatif de A et B serait inversé, c'est-à-dire que B serait plus récent que A. On décrit l'application des paléopôles de Thanet aux problèmes tectoniques grenvilliens et svéconorvégiens.

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Introduction

The Hastings Basin of southeastern Ontario forms a region of relatively low (greenschist) metamorphic grade within the generally higher-grade Grenville structural province of the Precambrian Canadian Shield. To the northwest, the grade of metamorphism increases to the upper amphibolite and granulite facies of the Hastings–Haliburton Highlands (Fig. 1). The entire region forms part of the Central Metasedimentary Belt (Wynne-Edwards 1972), composed largely of supracrustal rocks of the Lower Proterozoic Grenville Supergroup which lies unconformably on basement gneisses of Archean age. The greatest thickness of supracrustal rocks within the Grenville Province is found in the Hastings Basin.

Buchan and Dunlop (1973, 1976) studied the Haliburton intrusions of the Highlands and isolated three magnetic remanences. Each was believed to post-date regional metamorphism of the Grenville Province on account of the high temperatures (600–700°C) associated with upper amphibolite and granulite metamorphic terrains. Survival of any

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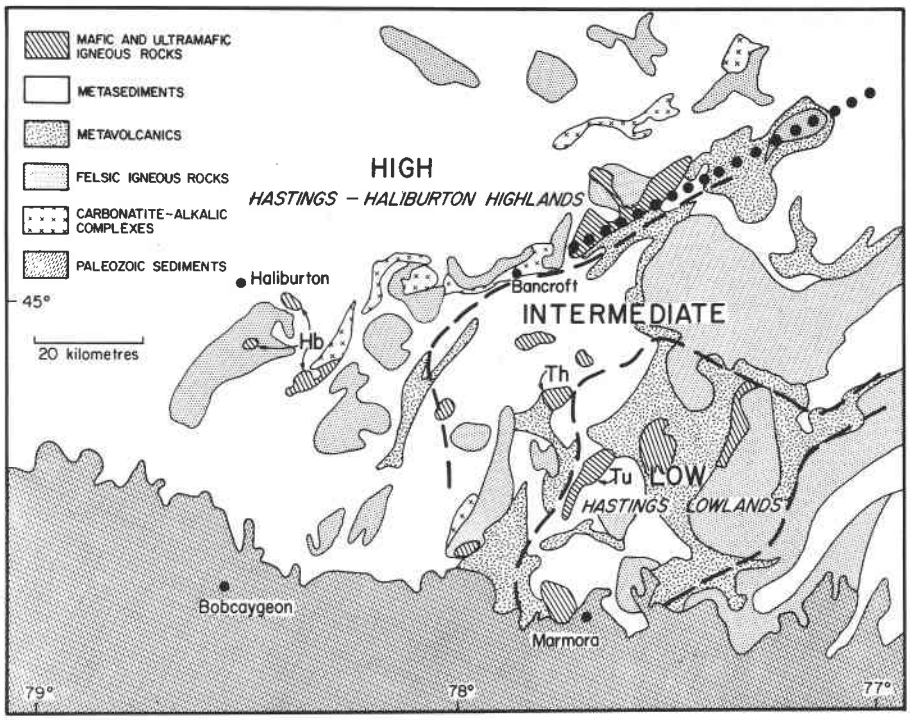


FIG. 1. Geology and metamorphic zones of the Hastings–Haliburton Highlands and Hastings Lowlands (after Lumbers 1964). The dashed lines represent boundaries between regions of high, intermediate, and low metamorphic grade. The symbols Hb, Th, and Tu designate the Haliburton intrusions, the Thanet complex, and the Tudor gabbro, respectively. The dotted line represents the approximate northern boundary of the suture zone proposed by Chappell *et al.* (1975).

pre-Grenville magnetic remanence is more likely to have occurred in the less intensely metamorphosed Hastings Basin. Hence, study of the Thanet gabbro complex on the edge of the Basin (see Fig. 1) is a logical extension of the work done in the Haliburton Highlands.

Geological Setting

The Thanet gabbro complex is located in Wollaston and Limerick Townships whose geology has been mapped by Hewitt (1962) and Lumbers (1969), respectively. It occupies a syncline in intermediate (amphibolite) grade terrain on the boundary with the low grade rocks of the Hastings Basins (Fig. 1). The body is bounded by marble and paraamphibolite on the north and south, by metavolcanics to the west, and metagraywacke to the east (Fig. 2).

The Thanet complex is composed largely of diorite, metagabbro, and gabbro. Hewitt (1962) considers the Thanet body to be a "multiple intrusion, with subsequent intrusions differing slightly in composition from one another." Three separate lithological intrusive units, each in contact with the others, are recognized by Lumbers (1969). Contact relationships suggest that all three were emplaced in rapid succession. Subsequently, the body was intruded by dikes and irregular masses of gabbro and diorite.

The age of the Thanet gabbro is uncertain. Hewitt (1962) and Lumbers (1969) agree that the Thanet body was most likely emplaced after the peak of regional metamorphism (the Grenville orogeny) in much of the Grenville Province. It be-

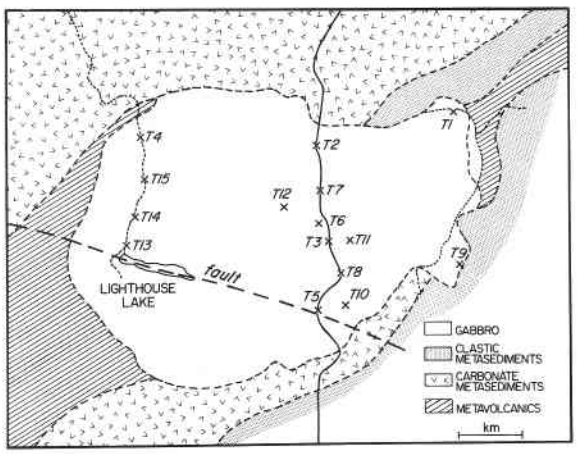


FIG. 2. Geology and sampling sites of the Thanet gabbro complex.

longs to a group of intrusions (Silver and Lumbers 1966; S. B. Lumbers, personal communication, 1978) whose U/Pb zircon ages are 1250 Ma or older. These bodies have subsequently undergone regional metamorphism (see Fig. 1). which occurred somewhat later in the Hastings area than elsewhere in the Grenville Province. Magnetic resetting outlined in the present study also indicated that the Thanet body has undergone a long and complicated thermal history.

Sampling and Measurement Procedures

Hood (1958), in a pioneering paleomagnetic study of the Hastings Basin, collected 15 samples from the Thanet gabbro. Nine samples were too weak to measure using the equipment then available. Specimens from the remaining six samples (all collected along the Old Hastings Road which bisects the body) were demagnetized in peak alternating fields of 125 Oe (1.0×10^4 A/m).

In the present study, sampling was extended throughout the body (Fig. 2). One hundred samples from 15 sites were oriented using a Brunton magnetic compass and cored in the laboratory. Three or more specimens from each sample were cleaned by stepwise alternating field (AF) demagnetization (8–12 steps) to 1000 Oe (8.0×10^4 A/m) or by stepwise thermal demagnetization (10–13 steps) to 600°C.

AF demagnetization was carried out successively along three perpendicular axes using a commercial demagnetizer. Thermal studies were conducted in a noninductive furnace whose temperature could be held within ±3°C for 1 h. Helmholtz coils were used to cancel the earth's magnetic field to 0.5%. Sample positions were reversed for each successive step in the thermal demagnetization process to ensure that no systematic remanence was introduced due to this residual field. Sample magnetizations were measured in a commercial spinner magnetometer.

Alternating Field Demagnetization Results

Forty-eight samples from 13 sites reach a common stable endpoint in AF's of 150 Oe (1.2×10^4 A/m) or less (Fig. 3). This steep downwards direction will be referred to as the A direction. Most coercivities are rather high (>400 Oe or 3.2×10^4 A/m). Intensities of NRM (natural remanent magnetization) exhibit a wide variation: near 1×10^{-6} emu/cc (1×10^{-3} A/m) at some sites (e.g., sample 57 of Fig. 3*a*); greater than 1000×10^{-6} emu/cc (1000×10^{-3} A/m) at others (e.g., sample 66 of Fig. 3*b*). These 48 samples have reverse magnetizations (positive inclinations) defining a south paleopole in the Pacific Ocean. An additional sample (36 from site T2) is normally magnetized (negative inclination). It also differs from other A samples in having no coercivities <1000 Oe (8.0×10^4 A/m).

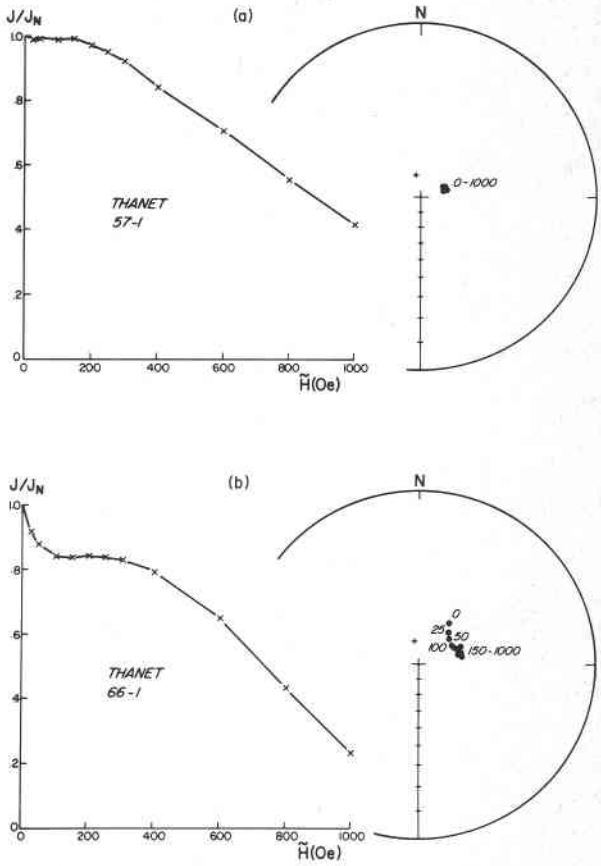


FIG. 3. The AF demagnetization characteristics for typical A samples. Directions are plotted on a stereographic (equal angle) projection. Closed circles indicate positive inclinations and open circles indicate negative inclinations. The direction of the present earth's field (with positive inclination) is shown as a cross.

At 8 sites there is evidence of a second stable component (hereafter called the B component) superimposed on the A remanence. Samples with a B component have coercivities which are, on the average, lower than those of stable A samples. Thus, as the B component is eliminated, the remanence direction follows a great circle path towards the direction observed in stable A samples (Fig. 4).

Such swinging behaviour is reminiscent of that observed in the Haliburton intrusions (Buchan and Dunlop 1973, 1976) and other Grenville studies (Fahrig *et al.* 1974; Irving *et al.* 1974*a*; Ueno *et al.* 1975). In the case of Haliburton rocks the direction of the softer (Hb_3) component was determined by vector subtraction between AF's of 150 and

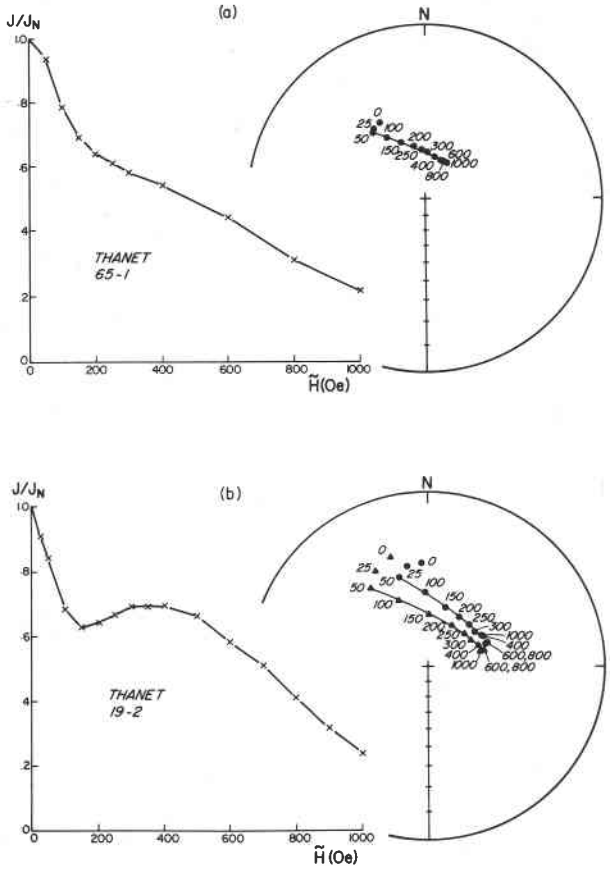


FIG. 4. The AF demagnetization characteristics for typical two-component (A + B) samples. Symbols are the same as in Fig. 3. Great circle paths have been superimposed on the directional data. In *b*, the directional changes for two specimens from the same sample are compared.

1000 Oe (1.2×10^4 and 8.0×10^4 A/m) since the underlying Hb_1 component was not demagnetized in this interval. However, the A component of the Thanet samples is not so hard (see, for example, Fig. 3), and it is difficult to select a range of coercivities which is appropriate for use of a simple vector subtraction technique.

An alternative method of determining a superimposed paleodirection involved determining the intersection point of converging great circles (Halls 1976). However, the directional scatter of Thanet B components is greater than that of A components, even for samples from a single site. Hence, great circle swings of the remanences of superimposed samples converge to the A direction (Fig. 5) and diverse towards the B direction. Thus, the great circle intersection technique can be utilized to determine a second and independent estimate of the A direction, but does not help define the B direction.

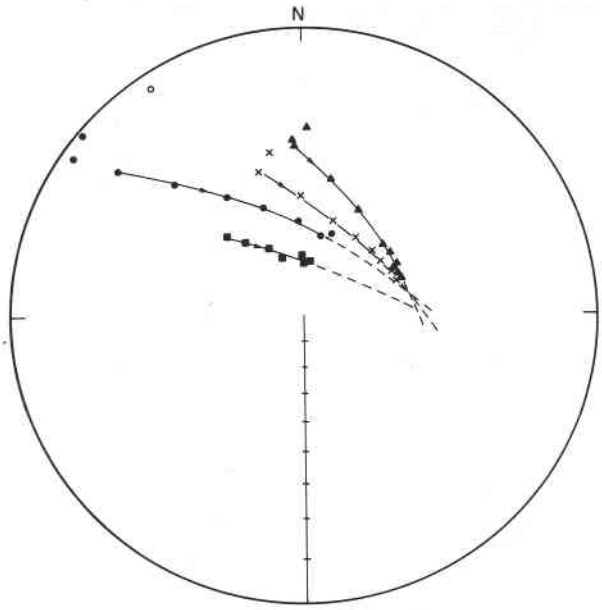


FIG. 5. Converging AF demagnetization paths for two-component samples from site 5. Great circle paths are superimposed on the data.

Thermal Demagnetization Results

Blocking temperature (T_B) spectra of A and B components were investigated by stepwise thermal demagnetization.

Stable A samples have T_B's confined to two narrow ranges: at some sites most blocking temperatures are 300 or 350°C (A_1 component of Fig. 6); other sites have T_B's of 525–560°C (A_2 component of Fig. 6). In general, low blocking temperature (A_1) samples are found to have low NRM intensities ($<100 \times 10^{-6}$ emu/cc or 100×10^{-3} A/m), whereas high blocking temperature (A_2) samples have much higher intensities ($>1000 \times 10^{-6}$ emu/cc or 1000×10^{-3} A/m).

Samples with a substantial B component have most of their blocking temperatures in the narrow range between 560 and 580°C (Fig. 6) — immediately above those of stable A_2 samples. In addition, many of these swinging samples (e.g., sample 42 of Fig. 6) exhibit a significant intensity increase in the 525–560°C range. Vector subtraction (see below) shows that this increase is due largely to the elimination of an A_2 component. This agrees with the observation that stable A samples at 6 of the 8 sites where swinging samples are found (sites T3–T5, T8, T12, T13) have A_2 characteristics (i.e., blocking temperatures between 525 and 560°C). At the remaining two B sites stable A samples have A_1 blocking temperature characteristics.

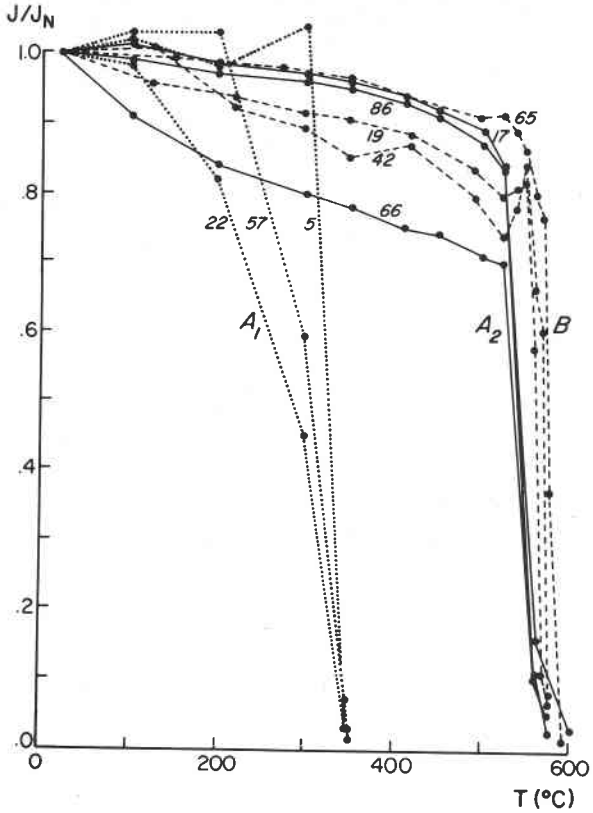


FIG. 6. Thermal demagnetization intensity characteristics for typical A_1, A_2, and B (or A_2 + B) samples.

Since the relative resistance of A and B components to thermal demagnetization is reversed to that observed for AF demagnetization, upon thermal demagnetization directions follow great circle paths away from the A direction, towards the B direction (Fig. 7). Usually, only minor directional and intensity changes are observed below 525°C (see Figs. 6, 7). These changes are believed to be associated with viscous components and can be readily eliminated by cleaning in peak AF's of 150 Oe (1.2×10^4 A/m). Above 525°C directions swing systematically to the 560 or 569°C step where they sometimes stabilize before becoming random at 577 or 591°C. This loss of directional stability coincides with the observed decrease in intensity in the 577 or 591°C step (Fig. 6).

Analysis and Statistics

A Component of Natural Remanent Magnetization

Of 13 sampling sites at which stable A samples were found, 3 have not been included in the statistical analysis since only a single sample at each did not contain a superimposed B component. At the other 10 sites $N \geq 3$. Normally magnetized sample 36 (site T2) has likewise been excluded from the statistical considerations. Paleodirections, paleopoles, and statistics are summarized in Table 1. The alternating fields which give the least scatter of specimen directions (lowest α_{95}, the radius of the circle of 95% confidence) define the sample direction. Sample directions are averaged to yield a site direction and sites are averaged to determine the intrusion mean.

Ten sites (45 samples) have a mean A paleodirection of $D = 83.6°, I = 58.6°$ ($k = 133, \alpha_{95} = 4.1$). The corresponding A paleopole lies at 165.3°E, 30.4°S. It should be noted that inclusion of the 3 sites with $N = 1$ does not significantly alter the paleodirection or related statistical parameters.

The narrow and widely separated ranges of blocking temperatures of A_1 and A_2 sites provide an ideal opportunity to determine if polar wander can be detected between the times when A_2 samples with T_B's >525°C and A_1 samples with T_B's < 350°C were blocked. Five of the 10 sites have T_B's > 525°C and define an A_2 paleodirection $D = 77.6°$, $I = 55.2°$ ($k = 151$, $\alpha_{95} = 6.2$) with corresponding paleopole 172.0°E, 32.3°S. The remaining 5 sites (T_B's < 350°C) define an A_1 direction $D = 90.9°$, $I = 61.7°$ ($k = 593$, $\alpha_{95} = 3.1$) and paleopole 158.4°E, 28.0°S. Site mean directions are plotted in Fig. 8*a*. The circles of 95 % confidence for A_1 and A_2 mean directions (Fig. 8*b*) just fail to intersect. Hence the difference between the A_2 and A_1 directions may represent polar wander during the interval of time during which that Thanet body cooled from >500°C to <350°C.

As noted earlier, an independent estimate of the A paleodirection can be obtained from the intersection point of great circles of swinging A + B samples following the method of Halls (1976). In the present analysis, only swings resulting from AF demagnetization were employed since they are usually much longer and hence more likely to better define great circles than are thermal demagnetization swings.

Fifteen such swings were of sufficient length to be usable. For each specimen, data exhibiting viscous effects at low AF's or the effects of instrumental noise at high AF's were eliminated. The remaining vectors in each case (see Fig. 9*a*) define a plane of remagnetization such that the sum of the squared deviations of the remagnetization vectors from the plane is minimized. Normals to each of the specimen's remagnetization planes are plotted in Fig. 9*b*. They likewise define a least-square plane whose normal (the square of Fig. 9*b*) gives the direction of

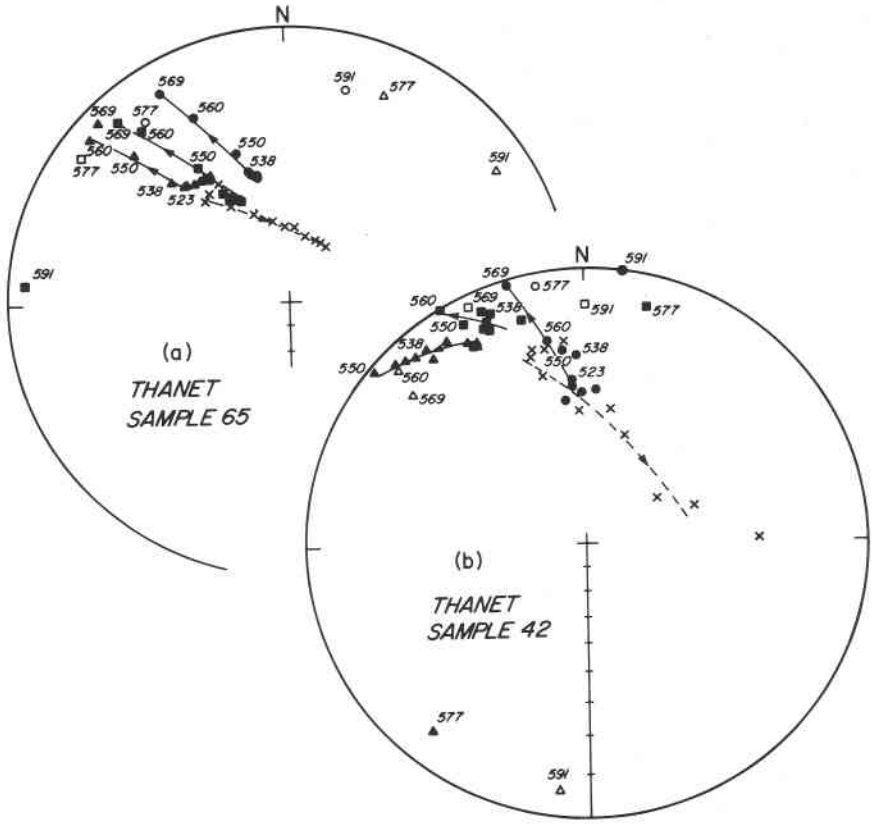


FIG. 7. Thermal demagnetization directional characteristics for typical two-component (A + B) samples. Triangles, squares, and circles designate directional changes for individual specimens after various peak temperatures (in °C). For comparison, AF demagnetization directions from a fourth specimen are plotted as crosses.

intersection of the great circles. Further details of the method are given in Halls (1976).

The direction of intersection (D = 73.2°, I = 55.3°) agrees closely with the mean direction for all A sites and in particular with the mean for A_2 sites. This might be expected since thermal demagnetization studies show that it is usually an A_2 component rather than an A_1 component which is superimposed with B remanences. Unfortunately, there is not a sufficiently large number of swinging samples to allow the intersection technique to be applied separately to A_1 + B samples and to A_2 + B samples.

The close agreement of the A direction obtained by the intersection method with that obtained using the endpoint method helps to confirm that no undetected B component resides in the A stable endpoint data. If such a component was present, it would tend to displace the endpoint data away from the intersection point along great circles towards the B direction.

B Component of Natural Remanent Magnetization

If we assume that the A component of swinging samples is essentially eliminated by 560 or 569°C, as in the case for all stable A samples which were studied (see Fig. 6), directions obtained just before swinging samples became demagnetized in 577 or 591°C should define the B endpoint.

Twenty samples swing away from the A direction along rough great circles upon thermal demagnetization. Eighteen of these samples from 8 sites have been used to determine a B paleopole for the Thanet gabbro. The remaining two samples are considered below.

At 6 of the 8 B sites, an A_2 component with blocking temperatures between 525 and 560°C (based on stable A samples found at the same sites) is superimposed on the B remanence (T_B's of 560–580°C). Hence, in order to study the great circle swings and better define the B endpoint, 3 or more specimens per sample were demagnetized at 525, 538, 550, 560, 569, 577, and 591°C as well as at

TABLE 1. Summary of A remanence directions† and paleopoles

Site‡	T_B range	N‡	Mean direction (deg) D‡	I‡	k‡	α_{95}‡
T2	A_1	5	87.3	62.7	142	6.4
T3	A_2	3	73.7	54.8	86	13.3
T4	A_2	4	87.3	49.2	54	12.5
T5*		1	90.9	59.1		
T6	A_1	5	86.1	58.3	45	11.5
T7	A_1	4	92.7	65.3	59	12.0
T8	A_2	3	66.5	51.9	63	15.6
T10*		1	83.6	55.1		
T11	A_1	6	95.3	62.1	101	6.7
T12	A_2	3	79.4	59.4	219	8.3
T13*		1	57.7	58.9		
T14	A_1	6	94.0	60.0	163	5.2
T15	A_2	6	81.3	59.8	127	5.9
A mean, unit wt. to sites ($N \geq 3$)		10	83.6	58.6	133	4.1
A paleopole			165.3°E	30.4°S	dp=3.6	dm=6.8
A_1 mean, unit wt. to sites ($N \geq 3$)		5	90.9	61.7	593	3.1
A_1 paleopole			158.4°E	28.0°S	dp=2.6	dm=5.8
A_2 mean, unit wt. to sites ($N \geq 3$)		5	77.6	55.2	151	6.2
A_2 paleopole			172.0°E	32.3°S	dp=5.8	dm=9.2

*Statistical parameters are not given for sites where $N < 3$.
†A remanence statistics are based on AF demagnetization data.
‡N is the number of units (sites or samples) averaged; D and I are declination and inclination; k is Fisher's precision parameter; and α_{95} is the radius of 95% confidence about the mean direction. Sites are the same as in Fig. 2.

more widely spaced intervals between 20 and 525°C.

Statistical analyses were carried out at follows. Sample directions (Fig. 10*a*; Table 2) were selected to correspond to the temperature at which specimen directions grouped most closely (lowest α_{95}). As might be expected on the basis of the above discussion, the closest grouping usually occurred at 560 or 569°C (after elimination of an A_2 component). At the 577 or 591°C step, the α_{95} invariably became very large as the sample was demagnetized, the B component lost, and the directions randomized.

Since the number of B samples at any given site varies from 1 to 5 (see Table 2), calculating an intrusion mean by giving unit weight to site means tends to grossly overweigh certain sites. However, for completeness, site statistics are included in Table 2. The mean Thanet B direction based on 8 sites (Fig. 10*b*) is $D = 307.7°$, $I = -11.9°$ ($k = 28.2$, $\alpha_{95} = 10.6$). Perhaps a statistical method which better represents the rather large scatter in the B data is simply to give unit weight to each of the B samples. Then the B mean direction becomes $D = 307.6°$, $I = -13.3°$ ($k = 9.6$, $\alpha_{95} = 11.7$) with paleopole 159.3°E, 20.3°N. Although this direction is almost identical to that obtained by averaging site means, the drastically reduced value of precision parameter, k, is probably more representative of the actual scatter in B directions evident in Fig. 10*a*. It might be noted that the rather large scatter of B sample means is reminiscent of a similar large scatter in B directions from other multicomponent Grenville studies (e.g., Haliburton, Mealy Mountain).

Since the B direction has been determined from swinging sample data, it is important to ensure that the directions used in the statistical calculations do indeed represent endpoints of the great circle swings. The problems which are encountered are examined in the following paragraphs.

Since the B component is usually erased in the 577°C step (occasionally 591°C), it is often difficult to be sure that a specimen's direction has actually reached a stable endpoint in the very narrow range between 560 and 577°C. This is evident in the examples shown in Fig. 7.

Even if stable endpoints are achieved, there remains the possibility that the blocking temperature spectra of the A and B components are identical

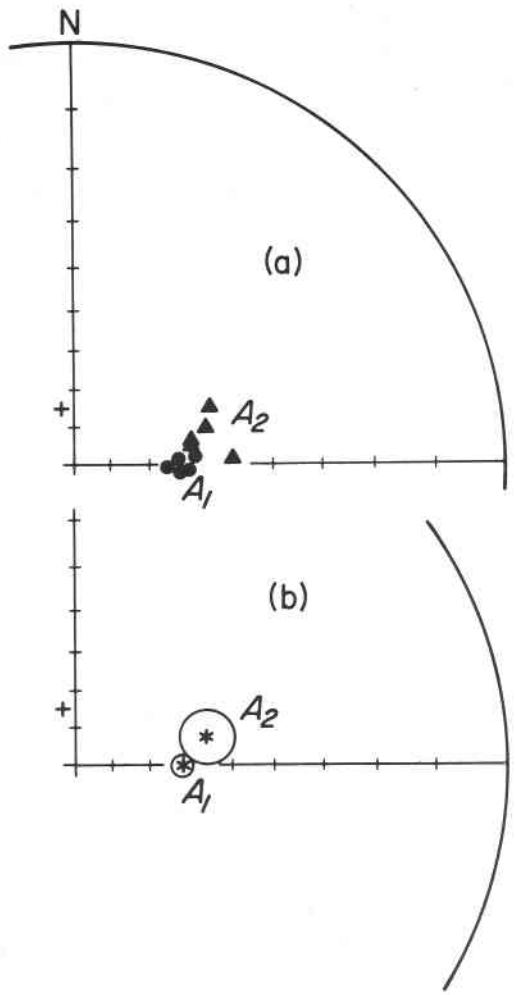


FIG. 8. (*a*) Mean directions for A sites ($N \geq 3$). A_1 site means are shown as circles, A_2 site means as triangles. (*b*) A_1 and A_2 intrusion means and their circles of 95% confidence.

above 560°C so that the great circle swings terminate without reaching a true B endpoint. Indeed, since one sample (36 of site T2), as previously noted, maintains a stable normal A direction and two further samples (73 and 74 of site T13) swing systematically from reversed A to normal A directions upon AF demagnetization, one might suspect that the true B component of swinging A + B samples is merely the reversal or near reversal of the A direction found at most sites and that thermal demagnetization has simply failed to totally remove the reversed A component. This argument, however, can be refuted in two ways. First, the three anomalous samples and no others are observed to have T_B's > 600°C, and hence do not belong to the same population as B samples whose blocking temperatures never exceed 580°C. Second, swinging samples 73 and 74 follow much more scattered paths on their way to the upper hemisphere than do B samples. This is not unexpected if 73 and 74 do, in fact, represent true reversals of the A direction since one would not expect to find a preferred great circle path if the directions are exactly opposite. On thermal demagnetization, for example, specimen 74-4 crosses the horizontal plane with a declination near 60°E. B swings, on the other hand, although somewhat scattered, invariably cross the horizontal plane in the northwest quadrant (see Fig. 9).

To further examine the possibility of contamination of the T_B range above 560°C by an A component, it is useful to analyse the component of magnetization being erased in this interval and compare it with other intervals. With this in mind, the components being eliminated in the intervals 525–560°C and 560–577°C were determined by vector subtraction. Directions of subtracted vectors in each case are plotted in Fig. 11. The scatter of 525–560°C directions in Fig. 11*a* is obviously smeared from the A direction towards the B direction suggesting that in some specimens, B blocking temperatures are <560°C. In the plot of Fig. 11*a*, 2 sites (T2 and T10) were eliminated because A blocking temperatures at these sites are, for the most part, below 350°C. The 560–577°C data are presented in Fig. 11*b*. Although the bulk of the subtracted vectors group near the B endpoint mean with roughly the same large scatter as the endpoints of Fig. 10, a few show a distinct displacement towards the A direction. Hence, at least a few samples appear to have some A blocking temperatures >560°C.

Origin of Multi-component Magnetism

Three distinct directions of remanent magnetization (A_1, A_2, and B) are present in the Thanet intrusion. It is important to consider their origins in order to establish their relative ages and thus determine the relative positions of their paleopoles in the Grenville polar wander path.

The simplest blocking temperature models consistent with observation are illustrated in Fig. 12. Stable A_1 samples have all T_B's below 350°C. Stable A_2 samples have all T_B's between 525 and 560°C. B samples have an additional blocking temperature range (560–580°C), carrying the B component, immediately above the A_2 range. The relative intensities of these two components determine whether the NRM of these swinging samples will be nearer the A or the B endpoint. Vector subtraction, described above, suggests that the B component overlaps the A_2 range in some samples (see streaking away from the A direction of Fig. 11*a*). Some slight A_2 contamination may also be present in the 560–580°C range (see Fig. 11*b*).

Some general conclusions concerning magnetic

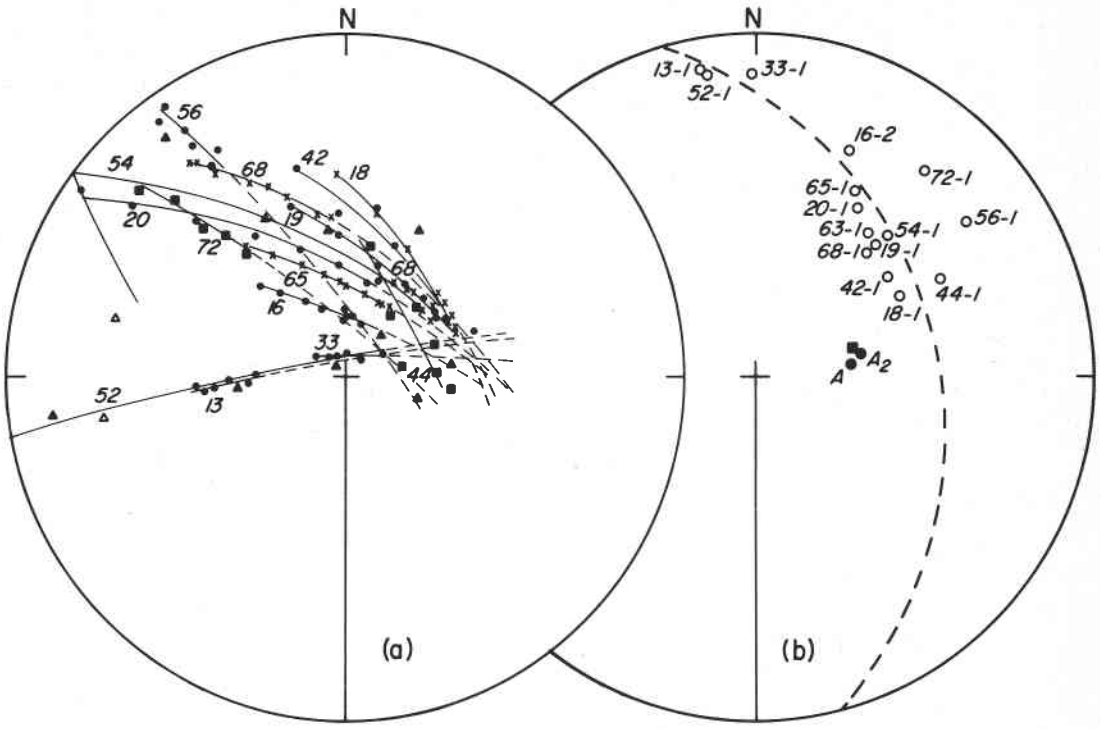


FIG. 9. (*a*) Converging AF demagnetization paths for two-component Thanet samples. Solid lines indicate segments of great circles along which data exist. These segments are extended as dashed lines. (*b*) Normals to remagnetization planes and least-squares fitted plane through them. The direction of the normal to this plane (the intersection point of the great circles) is shown as a solid square and compared with the mean A direction and mean A_2 direction.

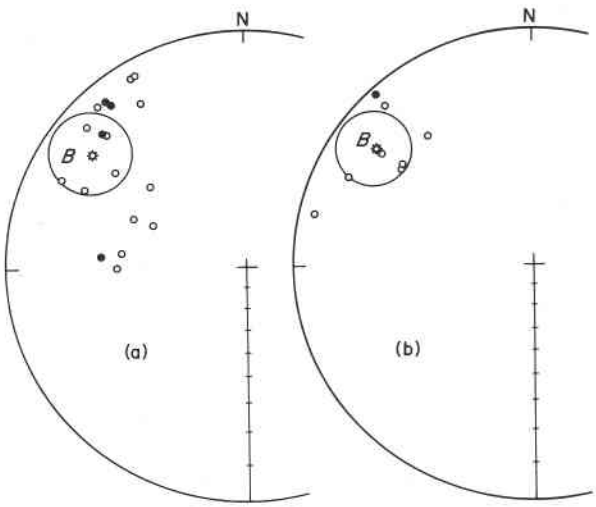


FIG 10. (*a*) B sample mean directions. Their mean (star) and its circle of 95% confidence are drawn. Directions are based on thermal demagnetization data (see text). (*b*) B site means and their mean and its circle of 95% confidence.

mineralogy can be drawn from preliminary examination of polished sections of four A_1, four A_2, and six B samples. Both A_2 and B components are likely carried by titanomagnetites which comprise the bulk of the opaque minerals in these samples. The carrier of the A_1 component may be either titanomagnetite (with a narrow range of blocking temperatures below 350°C) or pyrrhotite, both minerals being observed in polished sections. Due to the superposition of more than one component of magnetization in many of the samples detailed rock magnetic and petrological studies are needed to define the relationships between the various components and the magnetic minerals carrying them.

The simplest model of NRM acquisition is one in which all three Thanet components are thermoremanent magnetizations (TRM's) or viscous partial TRM's (VpTRM's), each acquired within its blocking temperature range during a single cooling event. The B component having the highest T_B's would be the oldest, the A_2 somewhat younger, and the low blocking temperature A_1 component the youngest. However, such a model is untenable since two adjacent blocking temperature ranges (A_2, B) have widely divergent directions of magnetization. Two cooling events spaced far enough apart to account for polar wander from B to A_2 are required. The second followed reheating at sufficiently high temperatures to unblock most grains with T_B's < 560°C but insufficient to reset those above 560°C.

TABLE 2. Summary of B remanence directions* and paleopoles

Sample	Site	N	Mean direction (deg) D	I	k	α_{95}
10	T3	2	318.4	−5.6	455	11.7
13	T4	3	274.7	28.2	21	27.7
15	T5	4	295.0	−44.3	204	6.7
16	T5	3	296.7	−16.8	315	6.9
18	T5	3	330.9	−4.7	47	18.1
19	T5	1	314.4	11.7	—	—
20	T5	4	296.8	−9.5	125	8.2
33	T2	3	269.6	−33.9	44	18.8
42	T8	3	330.6	−4.7	21	27.6
44	T8	3	310.9	−35.5	19	29.1
52	T10	4	276.3	−35.4	67	11.2
54	T10	3	314.7	−12.7	32	22.0
56	T10	3	328.4	−12.1	6	57.1
63	T12	3	321.4	7.6	6	54.5
65	T12	3	321.0	5.9	101	12.3
68	T12	3	312.5	−7.4	53	17.1
72	T13	3	307.2	−22.2	98	12.5
90	T4	3	293.7	−36.4	27	24.3
Mean B, unit wt. to samples		18	307.6	−13.3	9.6	11.7
B paleopole			159.3°E	20.3°N	dp=8.0	dm=11.3
	T2	1	296.8	−9.5	—	—
	T3	1	318.4	−5.6	—	—
	T4	2	283.7	−4.1	3	>90.0
	T5	5	307.4	−12.9	10	25.0
	T8	2	321.7	−20.3	10	87.9
	T10	3	308.5	−21.3	9	44.7
	T12	3	318.3	2.0	71	14.7
	T13	1	307.2	−22.0	—	—
Mean B, unit wt. to sites		8	307.7	−11.9	28.2	10.6
B paleopole			159.7°E	20.9°N	dp=7.3	dm=10.1

*B remanence statistics are based on thermal demagnetization data.

Theoretical calculations of Pullaiah *et al.* (1975) suggest that a temperature of ~540°C would be necessary to reset T_B's of 560°C in pure magnetite if the maximum reheating temperature were held for ~1 Ma. The second cooling must be slow enough to account for the small but measurable polar wander which is observed between acquisition of the A_2 and A_1 components.

In this model the B component would be considered as an early (perhaps primary) remanence which survived an ~540°C reheating event. The rough correspondence of the peak temperature (540°C) with peak temperatures necessary to produce the amphibolite grade metamorphism (~550°C according to Turner (1968)) of the Thanet region suggests that the reheating event which reset A_2 and A_1 blocking temperatures may be the regional metamorphism of this area (see Fig. 1). Hence, the B component may predate (i.e., see through) the regional metamorphism and was not erased because the reheating of the Thanet area (adjacent to the low metamorphic terrain of the Hastings Basin) was not as severe as in much of the Grenville Province.

Although the above model provides a simple self-consistent explanation of the acquisition of the A_1, A_2, and B remanences as TRM's (or VpTRM's), we must also consider the implications if one or more of these components was acquired as a chemical remanent magnetization (CRM). Even the above TRM model requires chemical magnetization to explain the normal A directions of samples 36, 73, and 74. Since their T_B's (often >600°C) are higher than those of the supposedly early B component, they must have been acquired as a low temperature phase (due to the production of hematite, perhaps?) below 560°C.

The B component itself could, at least in theory,

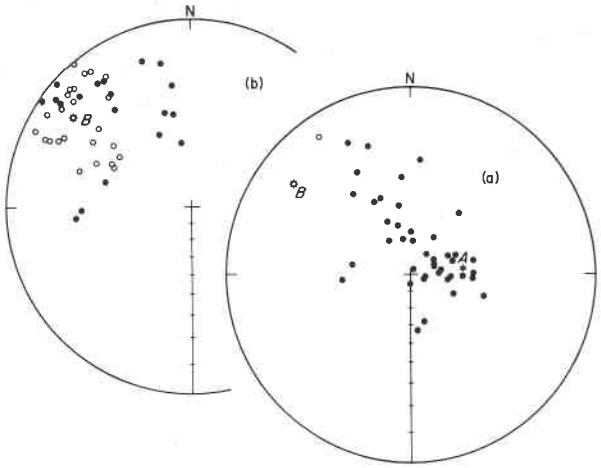


FIG. 11. (*a*) Directions of component erased upon thermal demagnetization between 525 and 560°C for specimens from A_2 + B sites. (*b*) Directions of component erased between 560 and 577°C for specimens from all A + B sites. A and B mean directions are shown as stars.

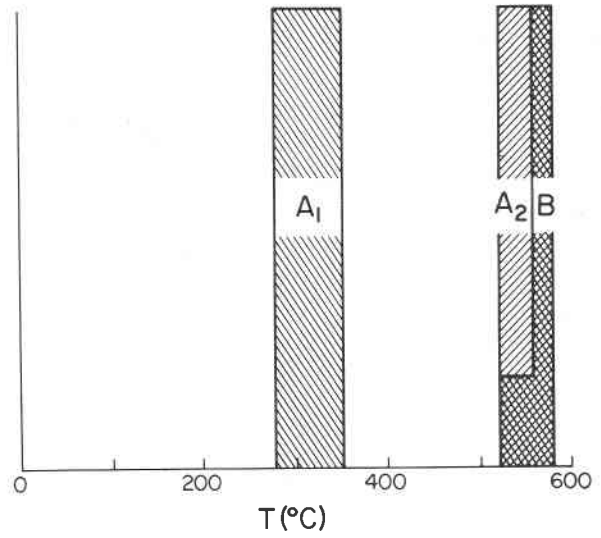


FIG. 12. Blocking temperature ranges of A_1 samples, A_2 samples, and two-component (A_2 + B) samples.

have been generated as a CRM at temperatures <300°C making it younger than both A components. The strongest argument against this model is that A_2 and B components of swinging samples (Fig. 12) have adjacent blocking temperature ranges and we must imagine these samples to have contained only grains with T_B's < 560°C (all magnetized in the A direction) until the B magnetic phase was generated with T_B's which, by coincidence, exactly filled the gap between 560 and 580°C, the Curie temperature of magnetite. This hypothesis cannot easily be ruled out, however, for stable Thanet A_2 samples do indeed have blocking temperatures up to but never exceeding 560°C. Generation of nearly pure magnetite as a low temperature chemical phase would complete the above picture.

Apparent Polar Wander and Tectonics

Grenville Paleopoles

Grenville paleopoles occupy a broad north–south band between southeast Australia and the north central Pacific (see Fig. 13*a*). Irving *et al.* (1974*b*), Irving and McGlynn (1976), and Buchan and Dunlop (1976) suggest that the paleopoles along this belt can be divided on the basis of magnetic properties into an A and a B grouping. The more southerly A poles (Hb_1, Me_1, Mr_1, Ws_1, Ws_1' from multi-component studies and Al, In, Mg, Ot, Wi from other highly metamorphosed bodies) have normal polarity (except Al and Ot which are mixed and Me_1 which is reversed), high coercivities, and high blocking temperatures (>550°C). B remanences (Hb_2, Hb_3, Mr_2, Ws_2 from multi-component studies and Fr, Gd, Gf from units believed to post-date Grenville metamorphism) with poles which lie further north are of reversed polarity (except normal Hb_3 and Me_2) and have lower coercivities and lower blocking temperatures (<400°C) than A remanences. Evidence presented by the above authors supports the hypothesis that the A remanences of multi-component studies pre-date the B remanences, resulting in a Grenville polar path from A to B.

One-plate or Two-plates?

Grenville paleopoles are displaced by as much as 50° from the polar wander path for the rest of North America (Interior Laurentia). Irving *et al.* (1972, 1974*b*), Palmer and Carmichael (1973), Buchan and Dunlop (1973, 1976), Stewart and Irving (1974), Ueno *et al.* (1975), and Irving and McGlynn (1976) have investigated the likelihood that this difference reflects the divergence of the Grenville Province and Interior Laurentia during the period when A remanences were acquired. Accordingly, B poles lying near the path for Interior Laurentia mark the convergence of a Grenville polar path (from A to B) with that for Interior Laurentia. This convergence corresponds to the suturing of the Grenville and Interior Laurentia landmasses.

Thanet paleopoles are plotted in Fig. 13*a*. The reverse Thanet A poles lie near but are displaced a few degrees east of the main Grenville A grouping—towards the reverse Me_1 pole from Mealy Mountain. On the other hand, the normal Thanet B pole falls close to the normal Haliburton

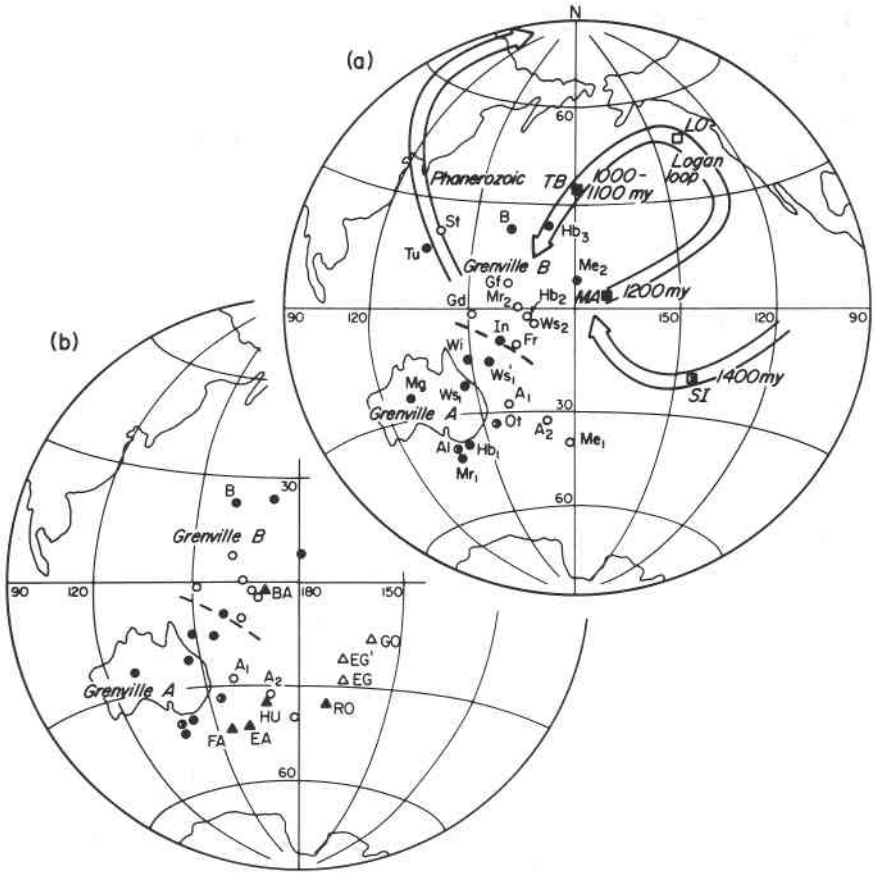


FIG. 13. (*a*) Grenville paleopoles and the North American polar wander path. Circles represent Grenville paleopoles: A_1, A_2, and B, Thanet gabbro complex (present study); Al, Allard Lake anorthosite (Hargraves and Burt 1967); Fr, Frontenac Axis dikes (Park and Irving 1972); Gd, Grenville diabase dikes (Murthy 1971); Gf, Grenville Front anorthosites (Palmer and Carmichael 1973); Hb, Haliburton intrusions (Buchan and Dunlop 1976); In, Indian Head anorthosite (Murthy and Rao 1976); Me, Mealy Mountain anorthosite (Fahrig *et al.* 1974); Mg, Magnetawan metasediments (McWilliams and Dunlop 1975); Mr, Morin anorthosite (Irving *et al.* 1974*a*); Ot, Ottawa basic intrusions (Irving *et al.* 1972); St, Steel Mountain anorthosite (Murthy and Rao 1976); Tu, Tudor gabbro (Palmer and Carmichael 1973); Wi, Wilberforce pyroxenite (Palmer and Carmichael 1973); Ws, Whitestone intrusions (Ueno *et al.* 1975). Squares represent tie points of the late Precambrian polar wander path for North America suggested by Robertson and Fahrig (1971) and Robertson (1973); LO, Logan sills; MA, Mackenzie dikes; SI, Sibley sediments; TB, Thunder Bay dikes. Closed symbols represent normal poles, open symbols reversed poles. The dashed line separates Grenville A poles from Grenville B poles as described in the text. (*b*) Sveconorwegian paleopoles (triangles) are compared with Grenville paleopoles (circles) after closure of the Atlantic. The Sveconorwegian paleopoles are: BA, hyperite (gabbroic) and amphibolite rocks of the Bamble and Kongsberg regions of south Norway (Poorter 1975); EA, Egersund anorthosite of Rogaland, south Norway (Hargraves and Fish 1972); EG and EG′, Egersund dolerite (diabase) dikes of Rogaland (Storetvedt and Gidskehaug 1968; Poorter 1972); FA, farsundite rocks of Rogaland (Murthy and Deutsch 1975); GO, Tuve dolerite dike near Goteborg, southwest Sweden (Abrahamsen 1974); HU, Hunnedalen dolerite dikes of south Norway (Poorter 1972); RO, Rogaland basement rocks, including the Egersund anorthosite (Poorter 1972).

B pole (Hb_3). Like other A and B poles, Thanet A's and B's have high and low coercivities, respectively. However, here the similarity ends, for the A remanence from the Thanet intrusion has lower blocking temperatures than does the B remanence. Therefore, on a simple blocking temperature model (see previous section) B would be older than A. This age sequence is incompatible with a convergent juncture in the polar wander path for North America. Rather, it fits one-plate models (Roy and Fahrig 1973; Fahrig *et al.* 1974; Stewart and Irving 1974; McWilliams and Dunlop 1975; Morris and Roy 1977) by incorporating Grenville data into a closed loop in the North American polar path. One might even suppose that the Thanet B pole predates the A grouping while other B poles post-date the A grouping. That is to say, the North American polar wander path loops through Grenville poles, recrossing itself in the area of B paleopoles.

Only a model of NRM acquisition in which the B

component of the Thanet body is a low temperature CRM (see previous section) would be compatible with a two-plate model. Hence, Thanet data offer no indisputable evidence to support a two-continent model.

Two-plate models also suffer from the lack of a convincing suture zone separating the Grenville and Interior Laurentia. Chappell *et al.* (1975) and Brown *et al.* (1975) on the basis of field study of the Elzevir granite batholith situated ~30 km east of the Thanet body present evidence of a broad northeast–southwest trending suture zone whose northern boundary lies along a line joining Bancroft with Renfrew to the northeast (see Fig. 1) and whose southern boundary is likely in New York State. Unfortunately, such a suture cannot be used to support two-continent models because many Grenville A paleopoles (Hb_1, Mr_1, Mg, Ws of Fig. 13*a*) which diverge significantly from the Interior Laurentia polar wander path come from sites located north of the proposed suture, that is, on the same side of the 'suture' as the Interior Laurentia results themselves. The Thanet results, coming from well within the proposed suture, provide further insight into this problem. Thanet A poles lie far to the south of the North American curve like the remainder of Grenville A results and thus show no drastic difference from poles north of the suture. Hence, the suture of Brown *et al.* (1975) cannot explain the 'Grenville problem' and must relate to an ocean closure which occurred prior to the magnetization of Grenville A type rocks.

Sveconorwegian Paleopoles

In addition to these local problems, the Thanet study provides insight into the relationship of Grenville paleopoles to those from the Sveconorwegian Province of the Baltic Precambrian Shield. The Sveconorwegian Province forms a wedge-shaped structural and radiometric zone in southeast Norway and southwest Sweden. Radiometric ages throughout the region are 1200–900 Ma (Kratz *et al.* 1968), similar to those found in the Grenville, and are believed due to thermal resetting during regional metamorphism. The similarities to the Grenville Province of the Canadian Shield are pronounced and become even more noticeable when the Atlantic Ocean is closed by rotation about a pole at 88.4°N, 27.7°E (following Bullard *et al.* 1965). It would appear that the Sveconorwegian Province was simply an extension of the Grenville Province prior to the overprinting of the Caledonian orogeny (in Norway, Britain, eastern coast of North America) and subsequent opening of the Atlantic (Wynne-Edwards and Hasan 1970).

Sveconorwegian paleopoles to 1975 are summarized and analyzed by Poorter (1975). In addition, a paleopole for farsundite rocks of Rogaland has recently been reported by Murthy and Deutsch (1975). After closure of the Atlantic Ocean about Bullard *et al.*'s (1965) pole of rotation, these poles can be compared with poles from the Grenville Province (Fig. 13*b*).

Ueno *et al.* (1975), on the basis of poles published prior to 1975, and Morris (1976) conclude that Sveconorwegian paleopoles are offset to the east of the Grenville grouping even after closure of the Atlantic about Bullard *et al.*'s (1965) pole of rotation. They suggest that a further movement of the Baltic Shield relative to North America, perhaps transcurrent motion, may be necessary to accommodate this difference. However, the Thanet A_2 paleopole of the present study also lies east of the main Grenville A grouping (as does the Mealy Mountain Me_1 pole). In fact, both the A_2 and Me_1 poles lie near or east of three Sveconorwegian poles (EA, FA, HU), whereas the Bamble–Kongsberg pole (BA) falls among Grenville B poles near the equator. The Egersund dikes (EG, EG'), the Tuve dike (GO), and the Rogaland basement (RO) poles, however, are displaced significantly east of the Thanet and Mealy Mountain poles.

Morris and Roy (1977) have recently proposed a 30° clockwise rotation of the Baltic Shield about a pole at 170°E, 40°S (after first correcting for closure of the Atlantic) to match poles from Sveconorwegian basement rocks and older intrusions with Grenville A poles and to match younger Sveconorwegian dikes with Grenville dikes (Gd of Fig. 13*a*).

Further Sveconorwegian paleopoles are needed before it will be possible to establish with certainty whether it is necessary to translate or rotate the Baltic Shield relative to the Grenville (after closure of the Atlantic).

Conclusions

The following conclusions can be drawn from the present study of the paleomagnetism of the Thanet complex.

1. Two directions of remanent magnetization coexist in the Thanet body, as evident from the great circle swings observed on AF demagnetization.

2. The A endpoint of these great circles, corresponding to the higher coercivities, can be isolated by AF demagnetization; however, a sufficient number of stable Thanet A samples are available, making the use of swinging samples unnecessary.

3. The intersection of great circle swings (after Halls 1976) gives an independent estimate of the A

direction. It agrees exceedingly well with the stable endpoint results.

4. The lower coercivity B endpoint can be isolated by thermal demagnetization, since the B component has higher blocking temperatures than the A. This was necessary due to a lack of stable Thanet B samples.

5. A small, but significant, amount of polar wander, apparently a function of the cooling history of the body, has been detected between the acquisition of the A_2 component (T_B's of 525–560°C) and the A_1 component (T_B's of 300–350°C).

6. In swinging samples, the A_2 and B blocking temperature ranges are adjacent, 525–560°C and 560–580°C, respectively, yet the directions of magnetization are far apart. This sudden change in direction cannot be accounted for on the basis of a single cooling episode unless the B component is a CRM. In this case, B would be younger than A. If both remanences are of thermal origin a second heating step which reset only grains with $T_B <$ 560°C in the A direction is necessary. This would imply A younger than B. It is not possible to eliminate one of these models with certainty at the present time. Detailed rock magnetic and petrological studies may aid in solving this problem.

7. The location of the Thanet body within the suture zone proposed by Chappell *et al.* (1975) and Brown *et al.* (1975) and the reasonable agreement of poles from the Thanet with other Grenville results from localities well to the north require that the suture pre-date the acquisition of Grenville remanences. It cannot explain the divergence of Grenville A poles from those for Interior Laurentia.

8. The displacement of the Thanet A_2 pole east of the main Grenville A grouping reduces the discrepancy (noted by Ueno *et al.* (1975)) between Grenville and Sveconorwegian data after closure of the Atlantic.

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